

Green Is Becoming the Baseline. Whole-Life Engineering Is the Next Frontier.

India has largely won the argument on operational energy. The harder questions - embodied carbon, construction and demolition waste, and what a building becomes at the end of its life - are engineering questions. Hybrid timber is one of the most practical answers available to us.



“Green technology in India's mega projects is no longer a differentiator reserved for premium developments - it is fast becoming the baseline expectation.”

MR. SANDEEP PINGALE

Founder & Managing Director, ECONSTRUCT DESIGN AND BUILD PVT. LTD.

01

The baseline has already moved

India's infrastructure story is now being written in solar panels, recycled materials and smart energy systems as much as in steel and concrete. As the country builds highways, metros, airports and industrial corridors at a scale few nations have attempted, sustainability has moved from a compliance checkbox to a design principle. The commercial case is plain enough: green buildings can cut greenhouse gas emissions by up to 35 per cent and trim maintenance costs by roughly 20 per cent - savings too significant for any large developer or public agency to ignore.

The scale is no longer experimental. India has the world's second-largest green building footprint after the United States, with the Indian Green Building Council having registered projects covering roughly 15.7 billion sq ft, of which more than 8,100 are already certified and fully operational. GRIHA, India's own rating system developed by TERI, is mandatory for all new central government building projects, while IGBC runs over 32 rating programmes spanning residential, commercial, industrial and infrastructure categories. Green standards are being embedded across the full spectrum of construction, not only in showcase towers.

Flagship projects have set the tone. Indira Paryavaran Bhawan in New Delhi - the Ministry of Environment's own headquarters - is India's first on-site net zero energy government building, holding both GRIHA 5-Star and LEED Platinum certifications, with a 930 kWp rooftop solar system large enough to meet the building's entire annual energy demand. When government buildings lead by example, private developers tend to follow.

Policy has reinforced the momentum. States such as Haryana, Rajasthan and Tamil Nadu now offer additional Floor Area Ratio, tax rebates and fast-tracked environmental clearances to IGBC- or GRIHA-certified projects, turning sustainability into a commercial advantage rather than an added cost. On the infrastructure side, the Mumbai-Ahmedabad bullet train corridor combines electric 2x25 kV

traction, Japanese J-Slab ballastless track, extensive elevated construction, noise barriers and green-certified station facilities. The Delhi-Mumbai Expressway incorporates wildlife overpasses and underpasses, tunnels through ecologically sensitive areas, solar lighting, rainwater harvesting and large-scale afforestation. Add 100 per cent wastewater reuse at Yashobhoomi and sustainable aviation fuel infrastructure at Navi Mumbai International Airport, and the pattern is unmistakable: sustainability is being engineered into these projects rather than applied to them afterwards.

That progress deserves to be celebrated. It also means the easiest gains are now largely behind us.

02

The question we have not yet answered

Almost everything we currently certify measures how a building performs while it is running. Very little measures what it cost to build, or what it will become when it stops.

That distinction is about to matter a great deal. As facades, glazing, automation and equipment improve and the electricity grid gets cleaner, the operational share of a building's lifetime carbon falls. What remains is the carbon locked into the material itself - cement, steel, glass, finishes - together with whatever is released when the building is eventually taken down. Quietly, the material decision becomes the carbon decision.

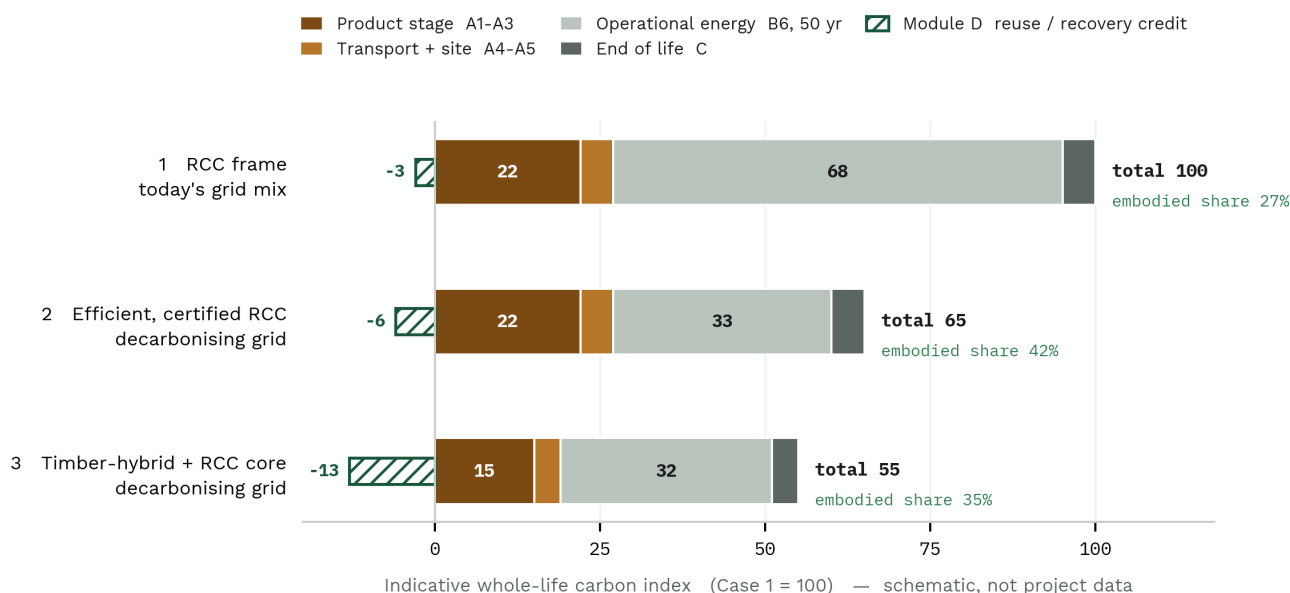


FIG 1 As operational energy is cleaned up, embodied carbon becomes the dominant remaining share - and the material specification becomes the main lever left to pull. Values are schematic and are shown to illustrate the shift in proportion, not to represent any specific project.

The same logic applies to waste. We rightly celebrate greener metros, airports, expressways and buildings. The more difficult question is what happens when today's buildings are renovated, redeveloped or demolished decades from now: where does the enormous mass of concrete, masonry, finishes, glass, steel and MEP waste actually go? India's rapid urban redevelopment cycle is generating construction and demolition waste faster than we are building the systems to absorb it.

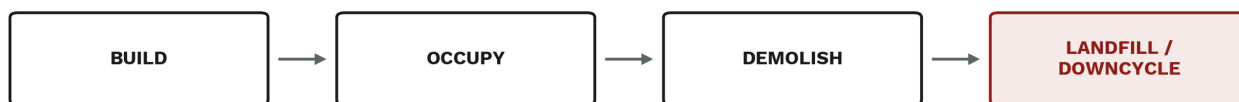
Future buildings should be engineered not only for construction and occupancy, but for repair, adaptation, disassembly, recovery and reuse.

03

Circular construction is a structural decision, not a recycling policy

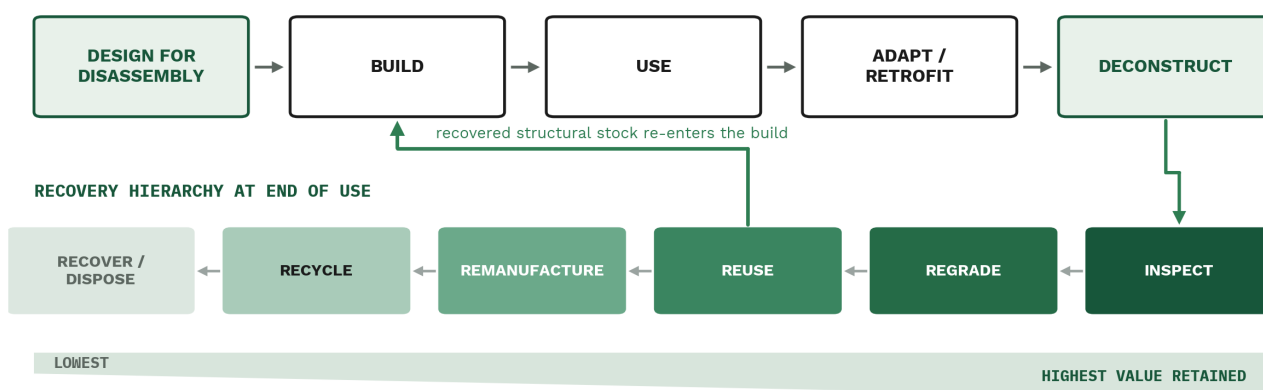
Circular construction changes the sequence from BUILD - DEMOLISH - DUMP to BUILD - USE - ADAPT - DECONSTRUCT - REUSE. In practice that sequence is decided at the design table and in the connection details, not at the demolition stage.

A. TODAY / LINEAR SEQUENCE



One-way flow. Embodied carbon and material value are written off in a single step.

B. CIRCULAR / DESIGNED FOR RECOVERY



The change is in the joints, not the aesthetics: mechanically fixed, demountable, documented.

FIG 2 The circular sequence is not a waste-management add-on. Whether a building can be taken apart is determined years earlier, by how its components are fixed together and how well they are documented.

Mechanically fixed panels, bolted connections, demountable facades, modular floor systems and accessible MEP zones allow a building to be adapted instead of prematurely demolished. At end of use, the recovery hierarchy should run inspect, regrade, reuse, remanufacture and recycle, with disposal as the last resort rather than the default. A digital material passport - what is in the building, where it sits, how it is fixed and what it is worth - turns that hierarchy from an aspiration into an inventory.

This is where the choice of structural material stops being a preference and becomes a strategy. Cast-in-place concrete is superb at a great many things; being taken apart is not one of them. Prefabricated, mechanically connected components are.

04

Where timber genuinely fits

Modern structural timber is not traditional wood construction. It covers glulam beams and columns, LVL, cross-laminated timber, structural wall and floor panels, timber-concrete composites and timber-steel or timber-RCC hybrid systems. These are manufactured, graded, quality-controlled engineering products, and they deserve to be assessed as such.

The honest case for timber is narrow and specific, which is precisely what makes it credible. For suitable building types, lower self-weight reduces foundation reactions and seismic inertia demand. Prefabrication compresses site programmes, reduces wet trades and cuts site waste. The floor plate -

the most repeated element in any housing project - is where those gains compound.

The case against blanket adoption is equally clear. Performance depends on connections, diaphragm continuity, moisture control, creep, vibration, fire and durability. Nothing about timber removes those obligations. It relocates them.

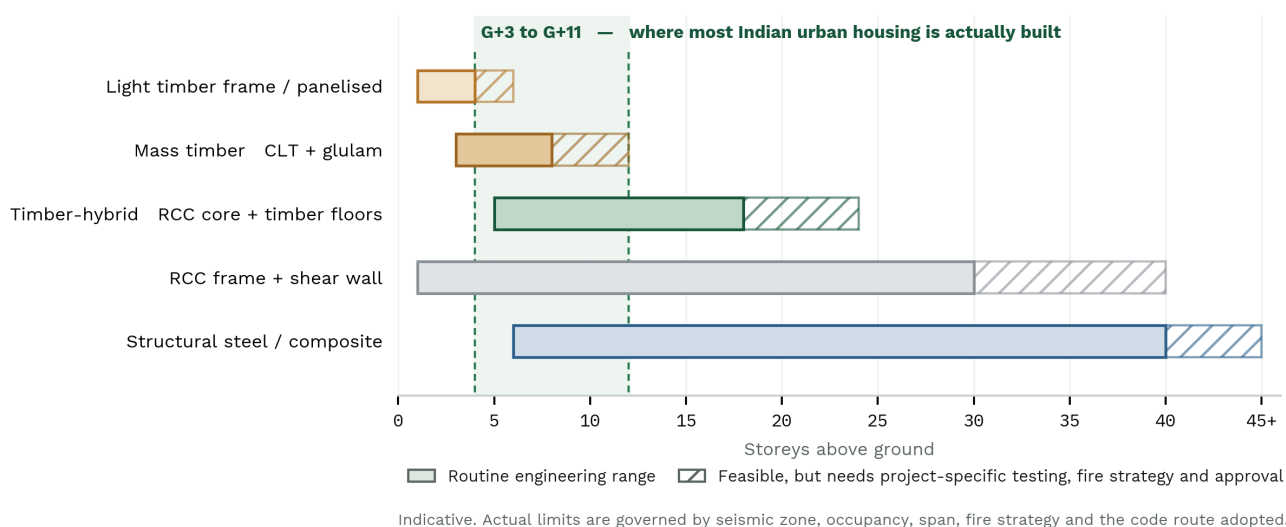


FIG 3 Timber is not a universal substitute, and it does not need to be. The hybrid band overlaps almost exactly with the mid-rise housing India actually builds - which is where the opportunity sits.

So the proposition is not that India should build in timber. It is that India should stop assuming a single material for every element of every building. Concrete where concrete performs best. Steel where steel performs best. Timber where timber performs best.

05

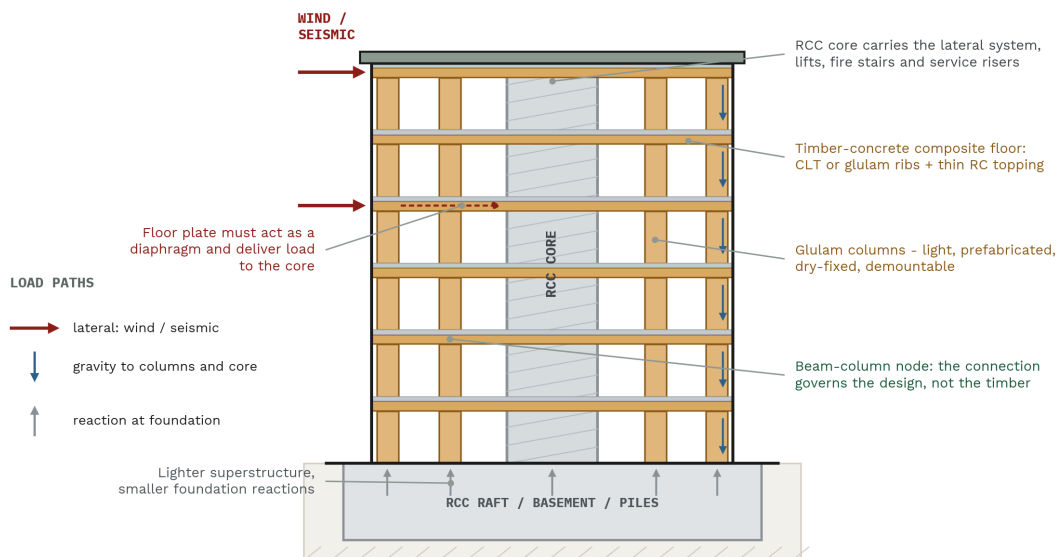
What a hybrid building actually looks like

In a mid-rise hybrid, the RCC core does what reinforced concrete does well: resisting lateral load, housing lifts, fire stairs and service risers, and providing a robust and familiar fire-fighting spine. The substructure and basement stay in concrete. Above that, glulam columns and CLT or timber-concrete composite floors deliver the repeating structure quickly and lightly, with a thin reinforced topping providing diaphragm action, acoustic mass and fire performance.

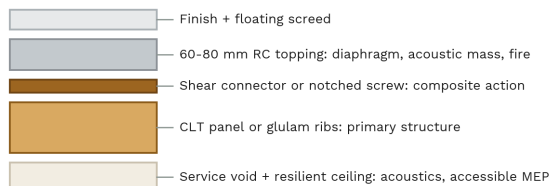
The engineering consequence is straightforward: lower seismic weight reduces design base shear and foundation reactions. But the governing checks shift with it. Connection stiffness and slip, diaphragm continuity, floor vibration under service loading, long-term creep, charring and residual section, moisture and termite protection become the design drivers. A hybrid building is not an easier building. It is a differently difficult one, and it must be resourced accordingly.

WHAT A HYBRID PROJECT HAS TO GET RIGHT

- Connection design, stiffness and slip - the real structural system
- Diaphragm continuity between timber floors and the RCC core
- Floor vibration and acoustic separation under service loading
- Charring rate, residual section and protected connections
- Moisture control during transport, erection and service life
- Termite and fungal protection detailed for Indian exposure
- Long-term creep and differential movement between materials
- Tolerance control between prefabricated and cast-in-place work



TYPICAL COMPOSITE FLOOR BUILD-UP



WHAT CHANGES IN THE CALCULATION

Lower self-weight reduces seismic inertia demand and foundation reactions. That is the gain.

The governing checks then move to connection stiffness and slip, diaphragm continuity, floor vibration, long-term creep, charring and residual section, moisture, termites and durability.

FIG 4 A hybrid is not a timber building with concrete in it. Each material is placed where its behaviour is an advantage, and the design effort shifts to the interfaces between them.

06

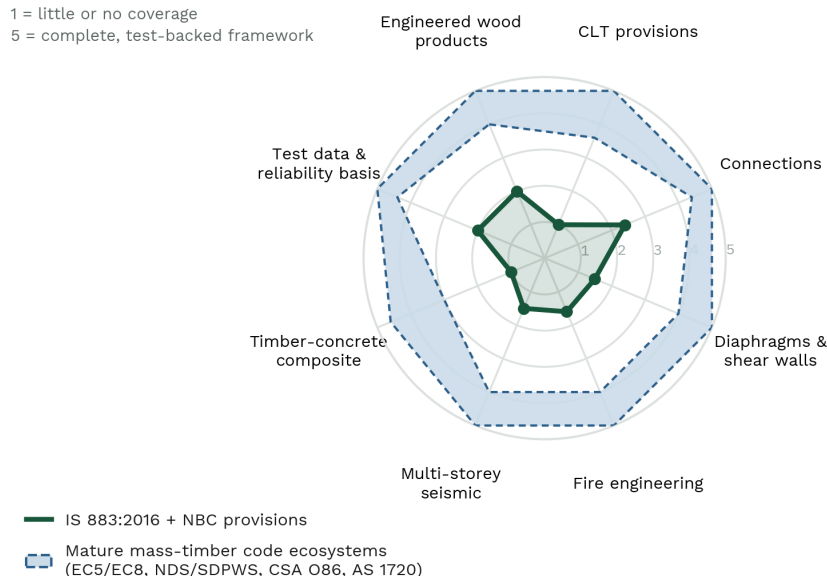
The code question is the real bottleneck

India already has IS 883:2016 for structural timber buildings, supported by NBC provisions and related standards. It is a sound conventional timber base. It was not written for a modern mass-timber sector.

The gaps sit at system level rather than in any single clause: engineered wood product standards, CLT provisions, advanced connections, diaphragms and shear walls, timber-concrete composites, fire engineering and multi-storey seismic behaviour - all of it supported by Indian test data on Indian species, manufactured in Indian conditions.

MATURITY FOR MODERN MASS-TIMBER DESIGN

1 = little or no coverage
5 = complete, test-backed framework



India has a sound conventional timber base. The gap sits at system level: products, connections, fire, diaphragms and multi-storey performance.

Overseas codes are benchmarks, not parts to be mixed clause by clause. Load factors, resistance factors and reliability assumptions belong to whole ecosystems.

Indicative assessment by ECONSTRUCT for engineered mass-timber design. Not a compliance rating.

FIG 5 The gap is not a matter of a few missing clauses. It is the absence of a complete, test-backed system for engineered mass timber - which is exactly what a national programme would need to build.

One caution matters more than any other. International codes should not be mixed clause by clause. Their load factors, resistance factors, material strengths and reliability assumptions belong to complete ecosystems - EC5 with EC8, NDS with SDPWS and ASCE, CSA O86 with NBCC, AS 1720 with AS 1170. Borrowing a favourable clause from one and a convenient factor from another produces a structure with no coherent reliability basis. Overseas standards are benchmarks to learn from while India develops its own provisions.

07

India cannot simply copy European details

Monsoon wetting, sustained humidity, coastal exposure, termites and fungi are not footnotes to the design; in much of India they are the design condition. Robust moisture control, drainage, ventilation, waterproofing and a genuine inspection regime are non-negotiable. Fire design must address charring rates and residual section, encapsulation, protected connections, sprinklers, compartmentation and cavity fire spread - and it must satisfy Indian approving authorities with Indian evidence.

Carbon claims need the same discipline. Trees absorb carbon as they grow, and timber can store biogenic carbon while it is in service. But "carbon negative" should never be assumed. A defensible lifecycle assessment covers plantation, harvesting, kiln drying, adhesives, fabrication, transport, erection, maintenance, reuse, recycling and final disposal. Anything less is marketing, and the industry will be judged harshly for it.

08

The farm end of the supply chain

The most interesting part of this opportunity is not urban at all. Scientifically managed farm forestry and agroforestry can create a domestic structural-timber supply chain without drawing on natural forests, and can diversify farm income while doing it. Agriculture, manufacturing and housing become one value chain rather than three separate sectors.

FROM FARM TO FRAME – AND BACK

Every stage carries a control gate. Miss one and the material is firewood, not structure.

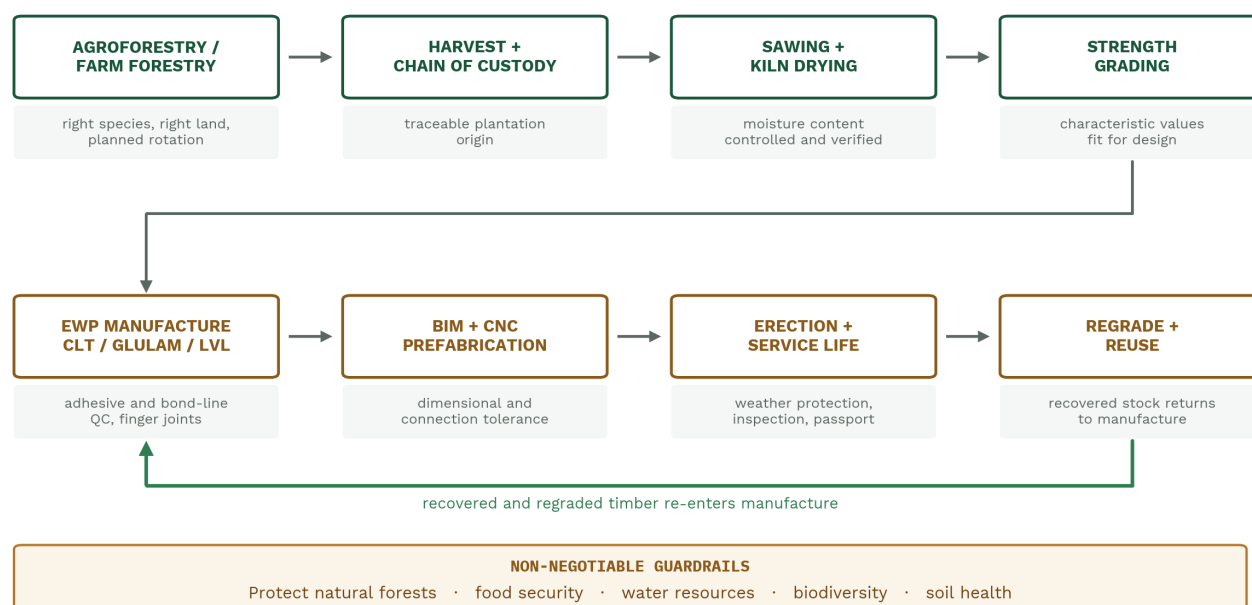


FIG 6 Structural timber is an agricultural product before it is a building product. Every stage between the plantation and the frame is a quality gate, and the chain is only as strong as the weakest one.

That only works if each stage carries a control gate: the right species on the right land with a planned rotation; traceable plantation origin; verified moisture content; strength grading that yields characteristic values a designer can actually use; adhesive and bond-line quality control in manufacture; dimensional and connection tolerance in prefabrication. Miss one and the material is firewood, not structure.

The guardrails are not negotiable either. Natural forests, food security, water resources, biodiversity and soil health come first. A timber industry that compromises any of them has defeated its own purpose.

09

What India should build next

None of this needs to wait for perfect conditions, but it does need to be organised. The programme is reasonably clear: structural species databases; engineered wood product standards; fire and seismic testing generating Indian data; demonstration housing that can be measured, instrumented and inspected over time; timber connection manuals; digital material passports; BIM and CNC prefabrication capability; and design-for-disassembly guidelines that apply to concrete and steel buildings too, not only timber ones.

The institutions already exist. The IITs and IISc, BIS, CSIR and the forestry institutes, agricultural universities, manufacturers and structural consultants each hold part of the answer. What is missing is the connective work of putting them on the same programme, with demonstration projects that are allowed to be measured honestly.

A NATIONAL PROGRAMME - TEN PRACTICAL STEPS

- 1 Structural species database for Indian timbers
- 2 Engineered wood product standards - CLT, glulam, LVL
- 3 Indian fire test data: charring, encapsulation, connections
- 4 Seismic and diaphragm testing for multi-storey systems
- 5 Timber connection design manuals for practising engineers
- 6 Instrumented demonstration housing, measured over years
- 7 Certified manufacturing with traceable plantation origin
- 8 BIM and CNC prefabrication capability in the supply chain
- 9 Digital material passports as a project deliverable
- 10 Design-for-disassembly guidance for all structural materials

10

The road ahead

The friction is real - certification costs, skill gaps, and slower adoption in Tier-2 cities. But the trajectory is not in doubt. Green technology is no longer a differentiator reserved for premium projects; it is becoming the baseline expectation. The next differentiator will be whole-life thinking: buildings designed for what happens after occupancy, made from materials chosen for what they cost the planet to produce and for what they are worth when they are recovered.

India should not build entirely in timber. It should build more intelligently. Hybrid structures can place RCC, steel and engineered timber where each gives the best combination of safety, economy, durability, speed and lifecycle value. The industry that adapts fastest to this shift will not simply be building greener. It will be building smarter, more resilient and more future-ready.

Right Material. Right Structure. Right Location. Right Lifecycle.

Mr. Sandeep Pingale is Founder and Managing Director of ECONSTRUCT DESIGN AND BUILD PVT. LTD. Diagrams prepared by ECONSTRUCT. Figures 1, 3 and 5 are indicative and schematic, intended to communicate engineering proportion and relative maturity rather than project-specific or code-compliance data.